

KEY INSIGHT AND RECOMMENDATIONS

SUPPLY CHAIN ANALYSIS

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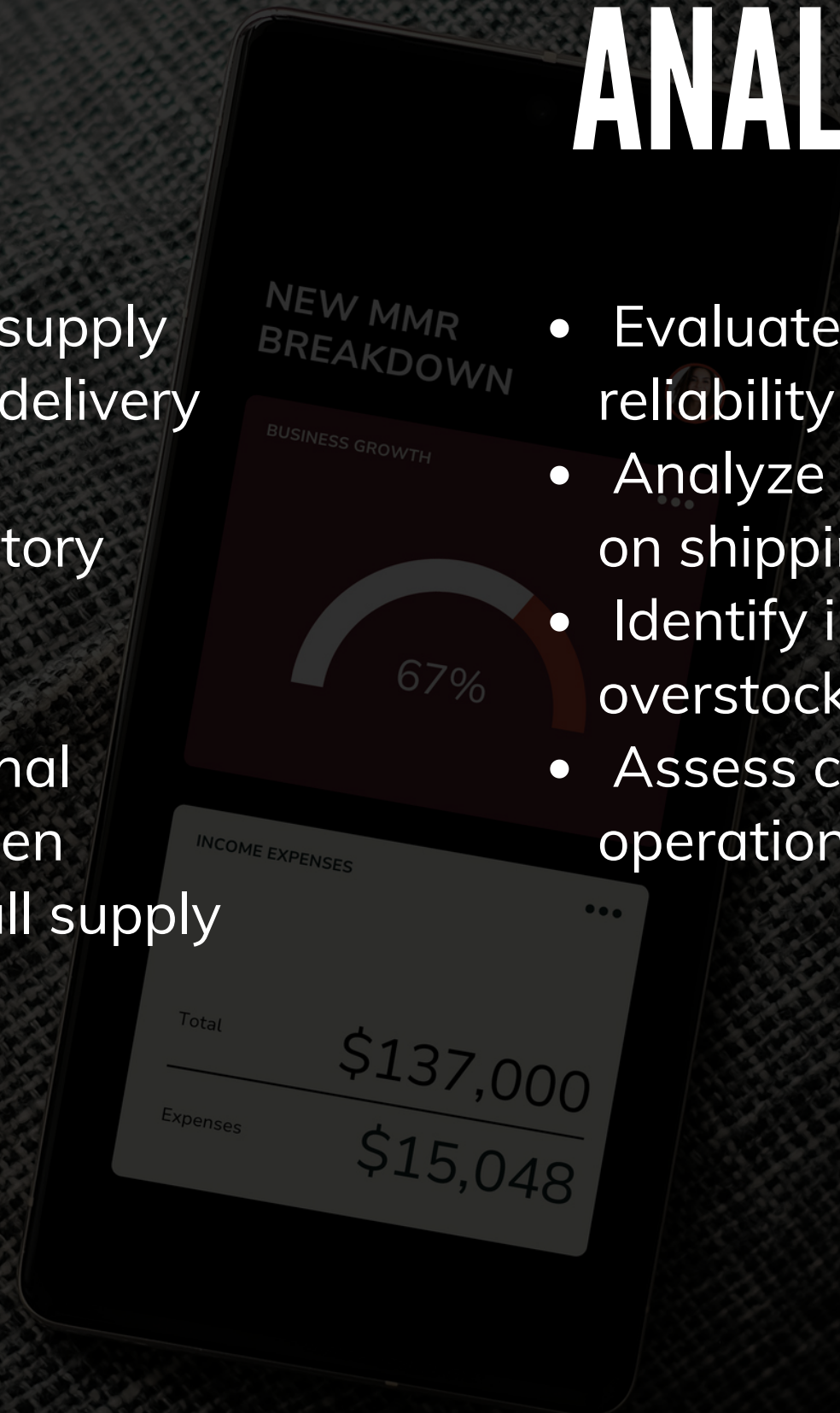
SUMMARY

This report presents an analysis of supply chain operations data, focusing on delivery performance, supplier reliability, transportation efficiency, and inventory management.

The objective is to identify operational inefficiencies and provide data-driven recommendations to improve overall supply chain performance.

ANALYSIS OBJECTIVES

- Evaluate supplier performance and delivery reliability
- Analyze the impact of transportation modes on shipping time
- Identify inventory risks such as stockouts and overstock
- Assess cost efficiency in relation to operational performance



DATA OVERVIEW

Operational Metrics:

- Shipping Time: Delivery duration (days)
- Total Cost: Overall logistics cost
- Total Lead Time: End-to-end process time

Supply Chain Variables:

- Supplier Name: Vendor providing products
- Transportation Mode: Road, Air, Rail, Sea
- Stock Levels & Demand: Inventory vs sales

Quality Metrics:

- Defect Rate: Product quality indicator
- Inspection Results: Pass / Fail status

DATA PREPARATION

Cleaned and standardized column names for consistency

Created key features:

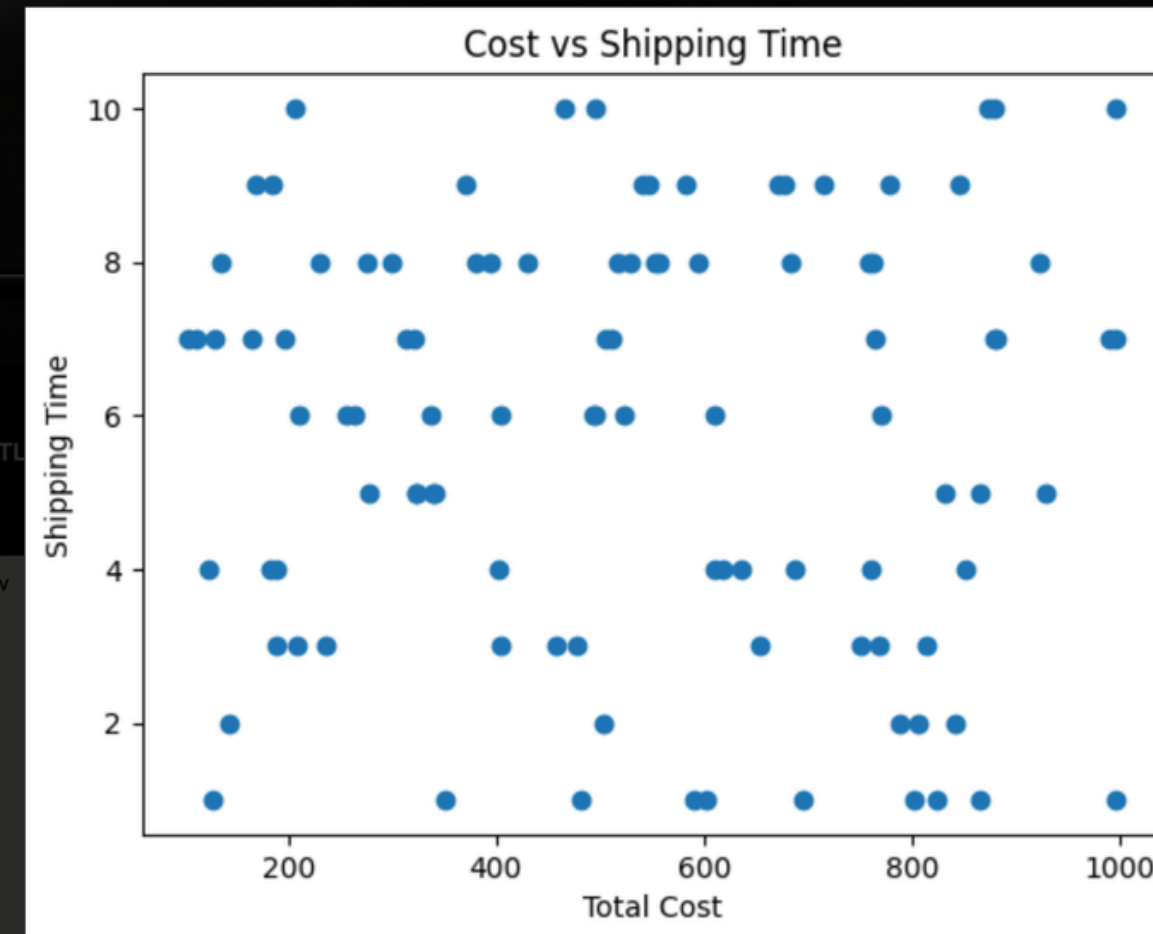
- Total Lead Time
- Late Delivery Indicator (~50% threshold)
- Stock Gap (inventory vs demand)

Ensured dataset completeness (no missing values)

KEY FINDINGS

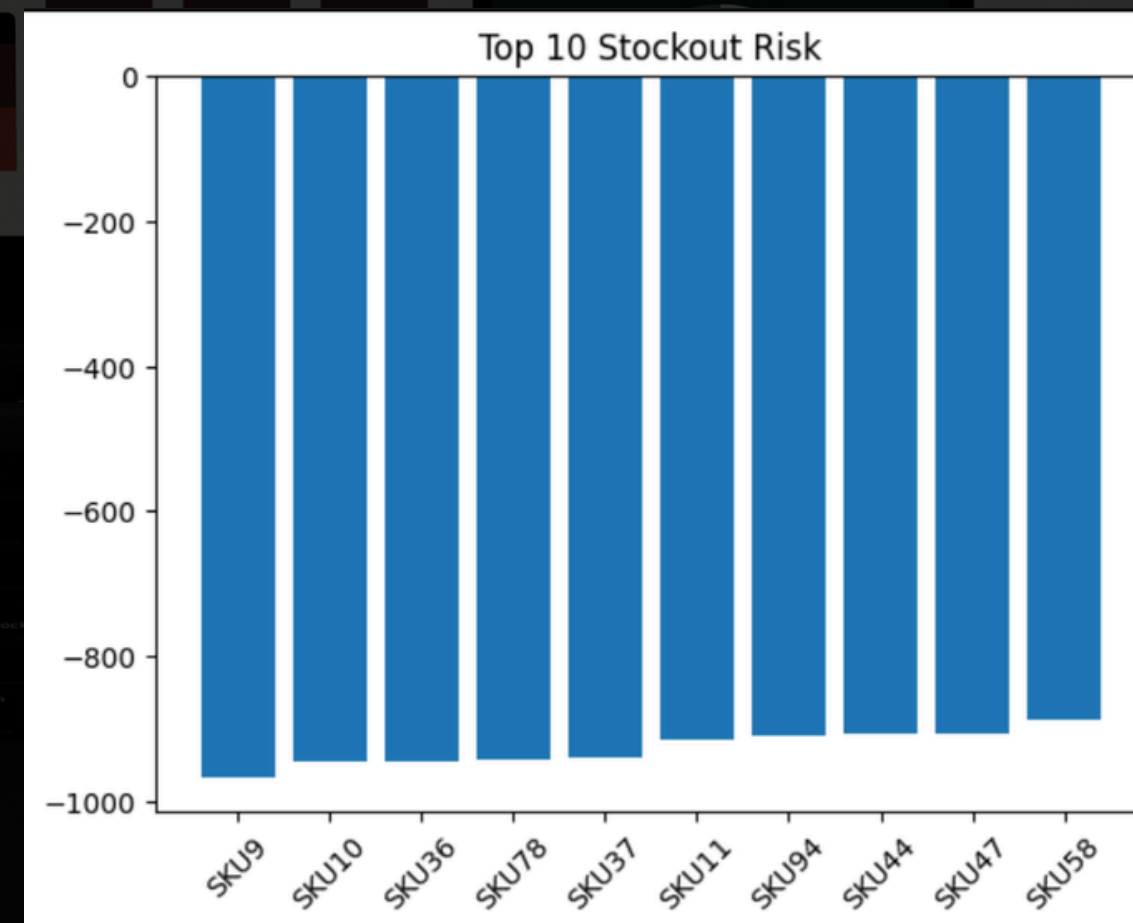
COST VS PERFORMANCE

Increasing logistics cost does not significantly reduce shipping time, indicating inefficient cost allocation across suppliers and transport modes.



INVENTORY RISK ANALYSIS

Top 10 SKUs show extreme stockout levels (up to ~1000 unit shortage), indicating severe demand-supply mismatch and poor inventory planning.



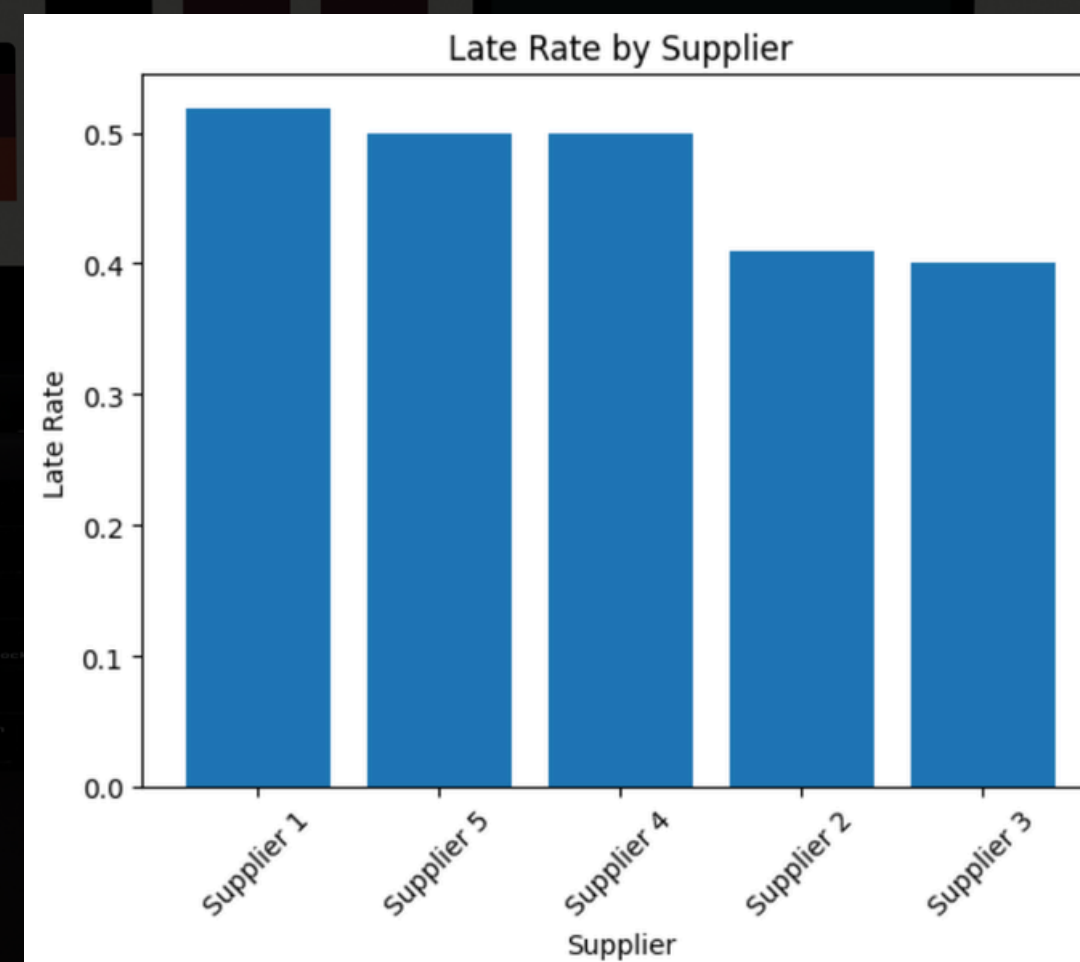
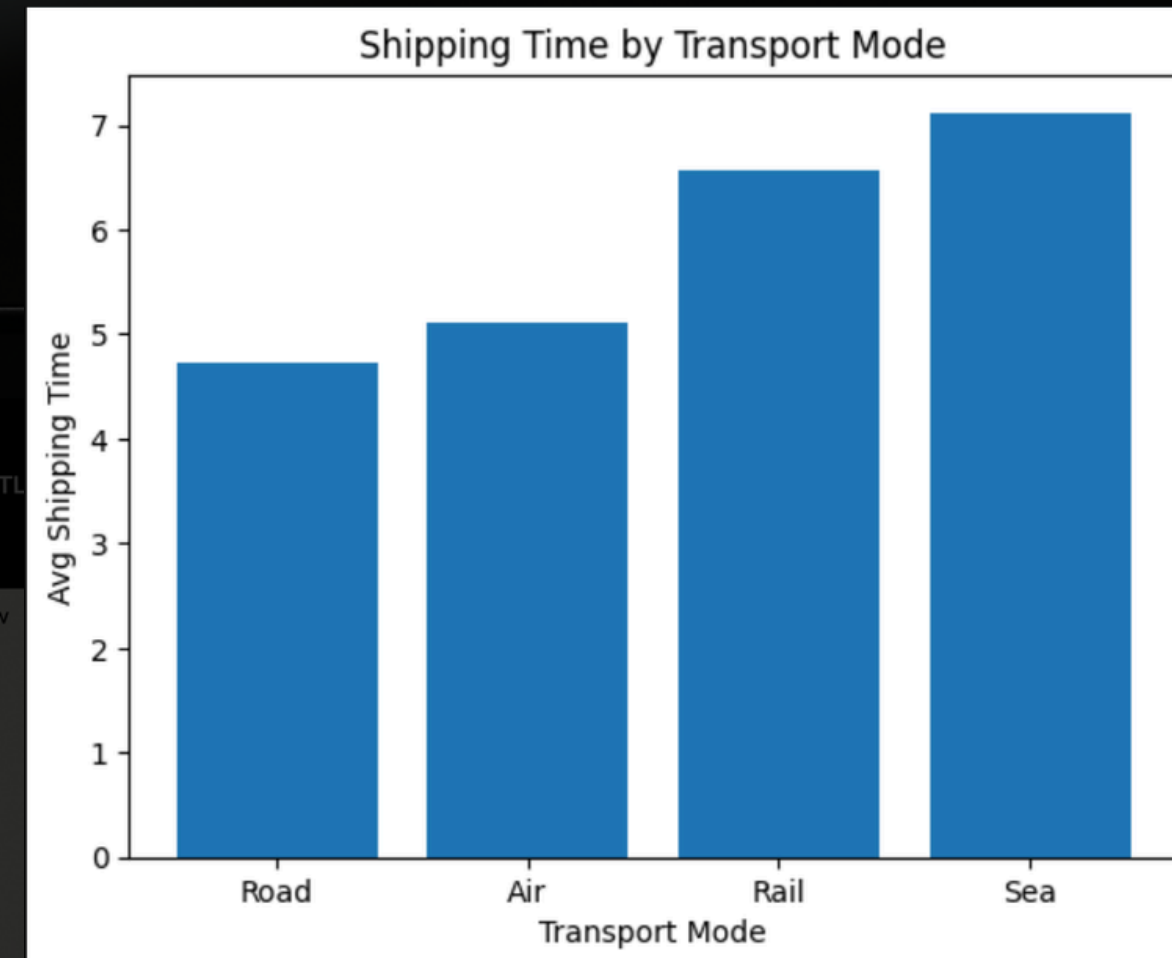
KEY FINDINGS

TRANSPORTATION ANALYSIS

Road transport is the fastest mode (~4.7 days), while sea transport is the slowest (~7.1 days), showing a ~50% increase in delivery time across transport options.

SUPPLIER PERFORMANCE

Supplier 1 has the highest late delivery rate (~52%), performing ~12% worse than the best-performing supplier, indicating a key operational bottleneck.



KEY INSIGHT

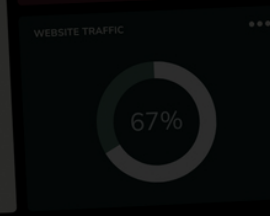
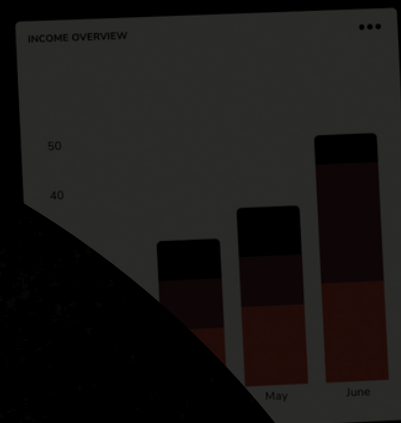
- No significant correlation between cost and delivery speed, indicating inefficient logistics spending
- Critical stockout issues identified, with top SKUs experiencing shortages of up to ~1000 units
- Transport mode significantly impacts delivery time, with sea transport being ~50% slower than road
- Supplier performance varies, with late delivery rates ranging from ~40% to ~52%, highlighting inconsistency in supplier reliability



RECOMMENDATIONS

- Rebalance supplier allocation by reducing reliance on high-delay vendors (>50% late rate) and shifting volume to consistently reliable suppliers
- Implement safety stock for high-demand SKUs and proactively address critical stockout risks (up to ~1000 unit shortages)
- Adopt a priority-based transport model: Road as default (fastest), Air for urgent demand, and Sea for low-priority shipments
- Eliminate high-cost, low-performance shipments and optimize logistics decisions based on cost-to-performance effectiveness

FINANCIAL OUTLOOK



FINANCIAL OUTLOOK

